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Entry

Serious Video Games: Tools for Learning, Training and Health

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Definition

Serious video games are digital games designed for purposes beyond entertainment, typically to support education, training, health interventions, or behaviour change. They combine game mechanics with psychological and pedagogical principles, such as feedback, repetition, goal-setting, and scaffolding, to create interactive environments that facilitate learning, skill development, and sustained engagement. In many cases, they are built to simulate realistic tasks or decision contexts, allowing users to practise skills, test strategies, and learn from consequences in a low-risk setting. Within cyberpsychology, serious video games are particularly valuable because they provide structured digital contexts for examining how technology shapes cognition, emotion, motivation, and behaviour. They enable researchers and practitioners to observe how users respond to digital rewards, challenges, social features, and immersive environments, as well as how these features influence outcomes such as self-efficacy, persistence, attention, and emotion regulation. As a result, serious video games operate at the intersection of psychological theory, human–technology interaction, and applied digital intervention design. This entry provides an overview of their development, theoretical foundations, applications, effectiveness, and associated ethical considerations.

Keywords: serious games; cyberpsychology; education; mental health; behaviour change; motivation; learning technologies

1. Introduction and History

Serious video games are digital games developed for purposes beyond entertainment, typically to support learning, training, health, or behaviour change across domains such as education, healthcare, defence, cybersecurity, and public policy. The term was introduced by Clark Abt in the 1970s to describe games designed with explicit educational or social objectives rather than amusement alone. Zyda later refined this definition, describing serious video games as a mental contest played according to specific rules that uses entertainment to advance training, education, health, public policy, and strategic communication goals [1].

Their interactive and rule-governed structure makes them particularly suited to contexts requiring skill development, decision-making, and applied problem solving. Within cyberpsychology, which examines how digital technologies shape cognition, emotion, and behaviour, serious video games provide structured environments in which motivation, learning processes, and emotional regulation can be both studied and applied. This entry situates serious video games within cyberpsychology, traces their historical development, outlines their psychological foundations, reviews key application areas, evaluates evidence of effectiveness, and considers ethical and future directions.

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2. Historical Development

Early examples of serious games predate digital technologies and were often grounded in structured role-based simulations. One historically influential example is the Prussian military war game Kriegsspiel, developed in the nineteenth century and used to train officers in strategic planning and tactical decision-making through rule-governed map simulations [2]. Such war-gaming practices formalised structured scenario-based learning and iterative feedback long before the advent of computers. Similarly, diplomatic and political role-play exercises, including early forms of Model United Nations simulations and mock congressional hearings have also been widely used in educational settings to develop negotiation skills, systems thinking, and perspective taking [3]. These analogue simulations reflected core principles that later became central to digital serious games, particularly experiential learning through active participation and reflection.

The theoretical foundations of such approaches were later articulated in experiential learning theory, which emphasises the importance of concrete experience, reflective observation, and iterative testing in skill acquisition [4]. These principles align closely with the mechanics of modern serious video games, where users engage in cycles of action, feedback, and adjustment within structured environments. With the expansion of computing technologies in the 1980s and 1990s, serious games increasingly moved into digital formats. Flight simulators and military training programmes became among the earliest large-scale implementations, allowing trainees to rehearse complex and high-risk procedures in safe and controlled environments [5]. These digital simulations demonstrated how interactive systems could replicate real-world tasks while enabling repeated practice without physical danger or resource cost.

Educational software also gained prominence during this period. Games such as The Oregon Trail introduced narrative-driven, decision-based learning into classrooms, allowing students to explore historical contexts through resource management and consequence-based gameplay. Research on game-based learning during the 1990s and early 2000s began to examine how interactivity and feedback mechanisms could support deeper engagement compared with passive instructional formats [6].

The early 2000s marked a further shift driven by improvements in graphical processing, broadband internet, and multiplayer connectivity. Online platforms enabled collaborative and competitive play, extending serious games beyond individual instruction toward socially mediated learning experiences. During this period, the concept of gamification emerged, referring to the application of game elements such as points, badges, and leaderboards in non-game contexts including corporate training, education, and health behaviour interventions [7]. Although gamification differs conceptually from full serious games, it reflects the broader diffusion of game mechanics into applied settings.

Serious games also expanded beyond formal training environments into scientific and civic domains. A prominent example is Foldit, an online protein-folding game that enabled non-experts to contribute to structural biology research, demonstrating that game interfaces could harness collective problem solving for scientific advancement [8]. Similarly, serious games have been developed to increase public understanding of environmental sustainability and climate policy, using interactive decision-making systems to model the consequences of complex policy choices.

More recently, advances in immersive technologies have transformed the field further. Virtual reality simulations are now widely used in surgical education [9], aviation training [10], and emergency response preparation [11], where immersive environments enhance procedural learning and situational awareness. The integration of artificial intelligence and adaptive systems has enabled serious games to personalise feedback and adjust difficulty dynamically, reflecting an increasing convergence of psychology, pedagogy, and

computational design. These developments illustrate how serious games have evolved from analogue simulation exercises into technologically sophisticated digital interventions spanning education, defence, health, science, and public engagement. Alongside this technological progression, serious video games have become increasingly relevant within cyberpsychology, as they provide controlled digital environments for examining how individuals interact with technology in relation to cognition, emotion, motivation, and behaviour. This has positioned serious games as both applied tools and research environments within the field.

3. Psychological Foundations

Serious video games are informed by a range of psychological theories concerned with learning, motivation, and emotional regulation within digital contexts. They typically incorporate meaningful choices, progressively increasing challenge, and opportunities for social interaction. These principles shape how games are designed, supporting engagement while also encouraging skill development and behaviour change.

Self-Determination Theory and Cognitive Evaluation Theory highlight the role of autonomy, competence, and relatedness in sustaining motivation [12,13]. Research on player engagement demonstrates that these needs are particularly relevant within digital gaming environments [14]. Flow theory further explains how engagement can be maintained when task difficulty is well matched to an individual's skill level [15]. Principles from behaviourism also underpin common game features, such as points, levels, and rewards, which reinforce desired behaviours through repeated feedback.

For example, language learning platforms such as Duolingo incorporate streak systems, progressive levels, and immediate corrective feedback, which reinforce competence and sustained engagement through structured reward schedules. Similarly, mathematics-based games such as DragonBox gradually increase task complexity while preserving player choice, supporting autonomy and mastery in line with Self-Determination Theory.

Learning theories provide an additional framework for understanding serious game design. Constructivist approaches emphasise active exploration, problem solving, and the importance of immediate feedback. Cognitive load theory focuses on managing the complexity of information to avoid overload while maintaining meaningful processing [16]. Repetition within game mechanics can support the consolidation of knowledge and the development of transferable skills. Virtual surgical training simulations [9] provide a good illustration of cognitive load principles, as tasks are typically scaffolded from basic procedural steps to full operative sequences, reducing extraneous load while maintaining germane cognitive processing. Likewise, science-based puzzle games such as Foldit [8] require iterative hypothesis testing, aligning with constructivist models of active problem solving and reflective adjustment.

Social and emotional perspectives are also important in understanding how serious video games function. Social Learning Theory suggests that observing others, including avatars or fellow players, can support the development of new skills and encourage prosocial behaviour [17]. In addition, serious games can contribute to resilience, emotional regulation, and self-efficacy by providing spaces where users can try different approaches, make mistakes, and learn from outcomes without real-world consequences and risks [18]. The Theory of Planned Behaviour also provides a useful framework for understanding how serious games may influence behaviour change, as they can shape attitudes, perceived social norms, and perceived behavioural control through interactive scenarios and feedback, thereby supporting intention formation and behavioural outcomes. For instance, cognitive behavioural therapy–based games such as SPARX allow adolescents to rehearse coping

strategies through avatar-guided challenges, strengthening emotional regulation and self-efficacy in a safe digital space.

4. Applications

Serious video games are used across a wide range of domains where interactive experiences can enhance learning, skill development, and behaviour change. In education, they support subjects such as science, mathematics, and languages through simulations, adaptive challenges, and feedback-rich environments. Tools such as DragonBox and Duolingo demonstrate how personalised feedback, mastery progression, and incremental difficulty can support sustained engagement and skill acquisition. For example, Duolingo uses spaced repetition, adaptive difficulty, and immediate corrective feedback to support language acquisition, while DragonBox introduces algebraic concepts through progressive problem solving that builds conceptual understanding without reliance on formal notation. Increasingly, these platforms also incorporate AI-driven personalisation, allowing content and feedback to adapt dynamically to user performance and learning patterns. Beyond individual subject learning, serious games are increasingly used to promote systems thinking and complex problem solving, particularly in STEM education. Research indicates that well-designed educational games can support deeper conceptual understanding when aligned with curricular goals and cognitive principles [19].

Serious games are also used in higher education and professional preparation. Virtual laboratory simulations allow students to conduct experiments in chemistry or biology without physical risk or resource limitations, while business and management simulations support experiential learning through strategic decision-making. These environments allow learners to test hypotheses, observe consequences, and refine strategies in iterative cycles, reinforcing constructivist principles of active learning.

In healthcare and mental health, serious games support physical rehabilitation [20], cognitive training [21], and psychological interventions [22]. Motion-controlled games, particularly those using sensor-based systems, have been used to aid motor recovery following stroke and neurological injury by encouraging repetitive, goal-directed movement in engaging contexts. These systems are often used in physiotherapy settings, where patient movements are tracked and tasks are adjusted in real time to support individual rehabilitation goals. Cognitive training games target memory, attention, and executive functioning, especially in ageing populations and individuals with mild cognitive impairment. Similarly, the game Re-Mission was developed to support young people undergoing cancer treatment, improving treatment adherence and disease-related knowledge by embedding health education within action-based gameplay [23,24].

Professional training represents another major application area. In military, defence, and emergency services, immersive simulations allow individuals to practise decision-making, coordination, and crisis response in high-risk scenarios without real-world consequences [25]. These simulations often integrate realistic time pressure and uncertainty, strengthening situational awareness and teamwork. In medical education, virtual surgery simulations have been shown to improve procedural accuracy and operating room performance [9]. Cybersecurity training increasingly uses gamified simulations and red-team/blue-team exercises to improve threat recognition, digital literacy, and incident response capabilities among professionals and the public. For example, cyber range environments simulate real-world network attacks, allowing users to practise identifying vulnerabilities, responding to threats, and managing system security under realistic conditions. Such platforms allow users to experiment with security protocols and observe the consequences of breaches in controlled digital environments.

Environmental and climate change education represents an emerging area for serious video games. Simulation-based environments enable players to examine the outcomes of decisions relating to ecosystems, energy consumption, economic trade-offs, and public policy. Decision-focused games, including those designed for climate strategy education, can support systems thinking and help users understand the interconnected nature of these challenges and the relationships between environmental, economic, and social systems. Research suggests that simulation-based climate education can enhance both knowledge and perceived self-efficacy regarding pro-environmental behaviour [26]. By modelling long-term consequences and policy trade-offs, serious games can make abstract global challenges more tangible and personally relevant.

Across these domains, serious games offer structured environments that combine engagement with applied learning, while enabling users to practise skills and decision-making in contexts that closely reflect real-world conditions. Examples of games referred to in this entry are listed in Appendix A. Their effectiveness, however, depends on alignment between design, psychological theory, and intended outcomes.

5. Ethical and Cultural Issues

Serious video games offer a range of advantages, including increased engagement, accessibility, and opportunities for personalised and scalable learning, but they also raise important ethical and cultural challenges that must be carefully considered. Their use across educational, healthcare, and professional contexts highlights both their potential as effective intervention tools and the importance of ensuring that design decisions are aligned with user well-being and equitable access. This dual role, as both learning environments and behaviour change tools, requires careful consideration of how psychological principles are applied in practice.

Accessibility is a key concern for neurodivergent individuals, those with language barriers, and users with varying levels of digital fluency. Games that assume uniform reading speed, attention span, motor control, or prior gaming experience may unintentionally disadvantage some users. For example, fast-paced interfaces or dense textual instructions may increase cognitive load for individuals with dyslexia or attention-related differences. Similarly, complex control systems may present barriers for users with motor impairments. Adjustable pacing, multimodal content, simplified navigation, and flexible feedback systems can help address these challenges and promote inclusive engagement [27].

Beyond accessibility, ethical issues also include privacy, data security, and informed consent. Serious games used in healthcare, psychological interventions, or educational assessment may collect sensitive personal data, including behavioural metrics, emotional responses, or performance analytics. As artificial intelligence and adaptive learning systems become more integrated into serious games, concerns around algorithmic bias and opaque decision-making processes become increasingly relevant. Bias embedded in training data or adaptive algorithms may unintentionally reinforce inequalities or disadvantage particular user groups if not carefully monitored [28]. Transparent data governance, explainable systems, and clear communication with users are therefore central to maintaining trust.

Cultural sensitivity is equally important. Games developed within specific social or cultural contexts may embed implicit assumptions about values, norms, or behaviours that do not generalise across populations. For example, leadership styles, communication norms, or health practices represented within a simulation may reflect Western frameworks that are not universally applicable. Without adaptation, such design choices may reduce relevance, engagement, or effectiveness in different regions, particularly where cultural norms around authority, collaboration, and communication differ significantly across societies [29].

There are additional concerns regarding manipulative design. Gamified reward systems and behavioural nudges, while potentially useful for engagement, can exploit psychological vulnerabilities if poorly implemented. Over-reliance on streaks, leaderboards, or variable reinforcement schedules may promote compulsive engagement or externalised motivation rather than deep learning or behaviour change [30]. The broader literature on persuasive technology highlights the ethical tension between supporting behaviour change and manipulating user behaviour, particularly when systems are intentionally designed to influence decision-making [31]. This raises important questions about autonomy, consent, and the boundaries of responsible design.

Addressing these issues requires user-centred, culturally aware, and transparent development processes, combined with ongoing evaluation. Ethical serious game design must balance engagement with autonomy, ensure inclusivity across diverse populations, and prioritise long-term well-being over short-term metrics of use.

6. Evidence of Effectiveness

Research indicates that serious games can support improvements in knowledge acquisition, procedural skills, and learner engagement, although outcomes are strongly influenced by design quality, instructional alignment, and implementation context. Systematic reviews in educational settings report gains in conceptual understanding and skill development when serious games are integrated with curricular objectives and guided instruction, but findings regarding motivation are more mixed [32]. In some cases, motivational benefits appear to reflect novelty effects rather than sustained intrinsic engagement.

In healthcare and professional training, serious games are associated with improved knowledge retention, procedural accuracy, and decision-making confidence, particularly when grounded in established psychological and pedagogical principles. Simulations that incorporate feedback loops, increasing complexity, and opportunities for reflection tend to demonstrate stronger outcomes than those relying solely on surface gamification features. In medical and emergency training contexts, immersive simulations have been shown to enhance performance in controlled assessments and improve skill acquisition in health professions education [33].

However, the durability and transferability of these effects remain key areas of concern. Evidence for long-term impact is comparatively limited, as many studies rely on short-term post-intervention assessments conducted immediately after gameplay [34]. Few investigations include follow-up measures extending beyond several months, and even fewer examine whether learning transfers effectively into real-world behavioural change or professional practice. This distinction between in-game performance and external application is critical for evaluating the true educational and behavioural value of serious games.

Some interventions provide promising evidence of sustained benefits. For example, the game Re-Mission, designed for young people undergoing cancer treatment, demonstrated improvements in treatment adherence and disease-related knowledge that were maintained over follow-up periods [23,24]. Nevertheless, systematic reviews consistently highlight methodological limitations across the field, including small sample sizes, heterogeneity in outcome measures, short intervention periods, and inconsistent reporting standards [35]. Overall, the literature suggests that serious games can be effective when carefully designed and theoretically grounded, but stronger longitudinal and ecologically valid research designs are needed to determine the extent to which benefits persist and generalise beyond controlled study environments. This limitation partly reflects the relative novelty of the field, particularly in relation to longitudinal and large-scale studies, which contributes to a more limited body of peer-reviewed evidence compared with more established areas of educational and psychological research.

7. Conclusions and Future Directions

Future developments in serious video games are likely to involve deeper integration of artificial intelligence and immersive technologies such as virtual and augmented reality. AI-driven systems can adapt content, pacing, and feedback in real time based on user performance, prior knowledge, and behavioural patterns. For example, adaptive learning platforms can modify task difficulty or provide targeted feedback based on user responses, creating more personalised and responsive learning environments. Virtual reality systems, in turn, can create realistic, embodied experiences that allow users to rehearse complex procedural, interpersonal, or high-risk tasks within controlled digital contexts. When grounded in psychological theory, these technologies have the potential to strengthen autonomy, competence, and engagement by maintaining optimal challenge and supporting sustained mastery experiences [36].

The integration of learning analytics and adaptive frameworks further expands this potential. By continuously monitoring performance indicators, behavioural data, and engagement metrics, serious games can dynamically adjust difficulty levels, scaffold tasks, and personalise learning pathways in ways that align with principles of cognitive load management and self-regulated learning [37]. Such adaptive systems may be particularly valuable for diverse and neurodiverse populations, where individual differences in pacing, attention, and feedback sensitivity require flexible design approaches. Emerging research suggests that AI-enhanced serious games may also support broader meta-skills, including self-regulation, resilience, and reflective decision-making, extending their relevance beyond discrete knowledge acquisition [38].

At the same time, these technological advances introduce new challenges. The increasing reliance on data-driven systems raises concerns regarding transparency, explainability, and algorithmic bias. Ensuring that adaptive systems do not unintentionally reinforce inequalities or disadvantage particular user groups remains a central ethical consideration. Additionally, while immersive Virtual Reality and AI-supported systems may enhance short-term engagement, questions persist regarding long-term effectiveness, transfer to real-world contexts, and sustained behavioural change.

Moving forward, the field would benefit from more rigorous longitudinal designs, standardised outcome measures, and clearer reporting frameworks to strengthen comparability across studies. Interdisciplinary collaboration between psychologists, educators, technologists, and ethicists will be essential to ensure that serious games remain theoretically grounded, empirically validated, and socially responsible. Taken together, this entry highlights the conceptual foundations, practical applications, and emerging challenges that shape the ongoing development of serious video games across domains. As digital technologies evolve, serious video games occupy a unique position at the intersection of behavioural science and interactive design, offering considerable promise provided that innovation is matched with critical evaluation and inclusive practice.

Serious video games therefore represent more than a technological innovation; they reflect a broader shift toward interactive, data-informed, and psychologically grounded approaches to learning and behaviour change. Their evolution illustrates how principles from motivation theory, cognitive science, and social psychology can be operationalised within digital environments that are scalable and adaptable across contexts. At the same time, their development highlights ongoing tensions between engagement and autonomy, innovation and evaluation, and personalisation and privacy. As serious video games become more embedded in educational systems, healthcare delivery, professional training, and public communication, the need for critical scrutiny increases. Future progress will depend not only on technological advancement but also on theoretical clarity, methodological rigour, and ethical responsibility. When carefully designed and transparently evaluated, serious

video games have the potential to contribute meaningfully to learning, health, and social outcomes. Their long-term impact, however, will depend on sustained interdisciplinary collaboration and a continued commitment to inclusive and evidence-based design.

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Appendix A. Example Serious Games Referred to in the Text

The following games are referred to in the main text:

- https://oregontrail.ws/the-oregon-trail-game/ (accessed on 1 April 2026)
- https://fold.it/ (accessed on 1 April 2026)
- https://ig.ft.com/climate-game/ (accessed on 1 April 2026)
- https://dragonbox.com/ (accessed on 1 April 2026)
- https://www.duolingo.com/ (accessed on 1 April 2026)
- https://www.sparx.org.nz/the-game (accessed on 1 April 2026)

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